



ICINGA

Open Source Monitoring

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Dublin | 5th June

TF-NOC Meeting



WWW.ICINGA.ORG

Agenda

- Introduction
- Tools and Platform
- Icinga vs. Nagios
- Architecture
- Icinga
 - Core
 - Classic UI
 - IDOUtils
 - Documentation

Agenda

- ▶ Icinga
 - ▶ Web
 - ▶ Mobile
 - ▶ Reporting
 - ▶ Addons & Plugins
 - ▶ Packages
- ▶ Roadmap
- ▶ What's next
- ▶ Questions & Answers

Introduction

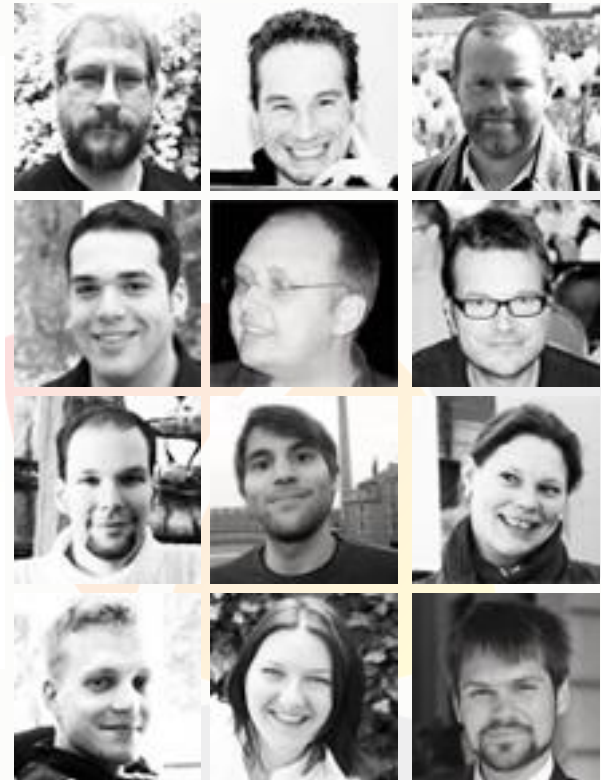
- ▶ Michael Friedrich
- ▶ 29 / m / Austria
- ▶ Developer & Administrator
 - ▶ University of Vienna, Computer Center
 - ▶ Internet Domain Administration
 - ▶ Department of ACOnet
- ▶ Lead Icinga Core Developer
- ▶ Icinga is ...
 - ▶ One of many projects at work
 - ▶ A hobby at home ☺

Introduction

Statistics

- ▶ Forked from Nagios in April 2009
- ▶ Over 100.000 downloads until now – more to come with 1.7 😊
- ▶ Different teams with independent responsibilities
 - ▶▶ Icinga Core
 - ▶▶ Icinga Web
 - ▶▶ Icinga Reporting
 - ▶▶ Icinga Marketing
 - ▶▶ Icinga Q&A
- ▶ 27 “active” members on the team

Team



Tools and Platform

Icinga Quality, Testing and Community Support

Website and Open Source Ticketing System

Icinga Core

C based source
MySQL, PostgreSQL and Oracle

Classic UI

IDOUTils

NRPE and NSCA

Icinga Web

based on PHP
using Sencha,
Agavi MVC

Mobile

Icinga Doc

based on
Docbook in
English and
German

Icinga Reports

based on
Jasper Reports

Tools and Platform

- ▶ Development Tracker based on Redmine
 - ▶ <https://dev.icinga.org>
- ▶ Wiki based on Confluence OSS license
 - ▶ <https://wiki.icinga.org>
- ▶ Build server based on Jenkins
 - ▶ Testing & Nightly Builds
 - ▶ <http://build.icinga.org/snapshots/>
- ▶ Website based on Wordpress & Plugins
- ▶ Support via Mail, Forum, IRC, Twitter, Facebook, ...

Icinga vs. Nagios

- ▶ Why a fork?
 - ▶ Single Nagios Developer
 - ▶ Community patches never applied
 - ▶ Stuck progress, no feedback
- ▶ Ideas behind the fork
 - ▶ Open up for the community
 - ▶ Create a stable architecture
 - ▶ Actively apply & review patches
 - ▶ Implement our features & bugfixes
 - ▶ Public Development Roadmap

Icinga vs. Nagios

- ▶ Problems with the fork
 - ▶ Stay compatible with Nagios Addons & Plugins
 - ▶ Review valuable Nagios Patches, integrate them
 - ▶ Invest time into comparison charts
 - ▶ Nagios Devs don't like Icinga Devs
 - ▶▶ Icinga Patches sent to Nagios get denied their origin
 - ▶ Nagios LLC spreads FUD about Icinga
 - ▶▶ Icinga is not Netways, they „only“ support the project
 - ▶▶ Icinga does not violate any IP or trademark
 - ▶ Community Portals threatened on Nagios & Icinga Topics
 - ▶▶ nagios-fr.org, nagios-portal.org
 - ▶▶ NOW monitoring-fr.org & monitoring-portal.org

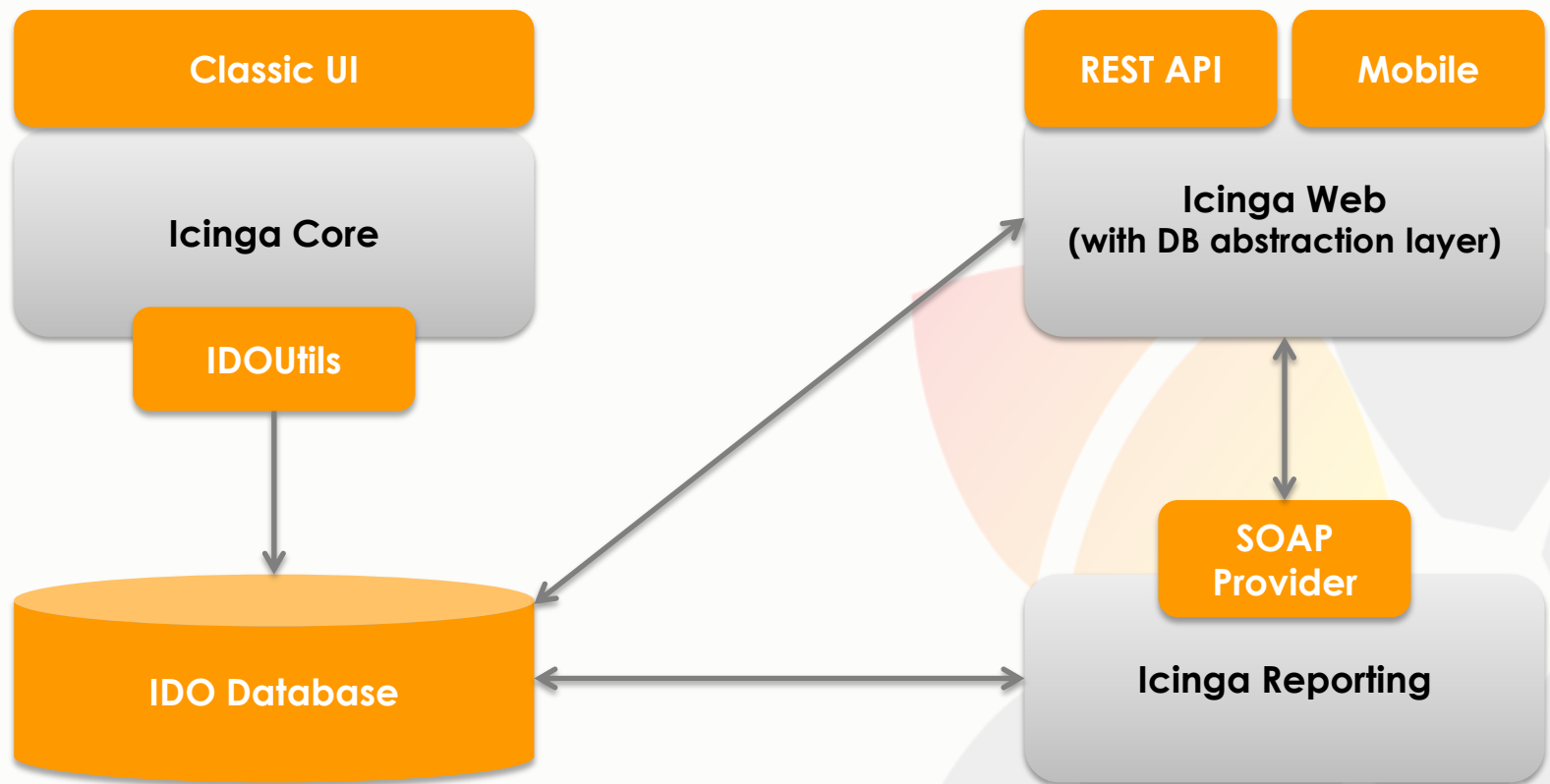
Icinga vs. Nagios

- ▶ Optimized database support
 - ▶ Better performance & platform flexibility
 - ▶▶ MySQL
 - ▶▶ PostgreSQL
 - ▶▶ Oracle
- ▶ 80+ bug fixes missing in Nagios ®
(<https://wiki.icinga.org/display/Dev/Bug+and+Feature+Comparison>)
- ▶ Many improvements in Core, Classic UI and a new web interface
 - ▶ IPv6 and IPv4 support, compound commands, log file search and logging
- ▶ Reporting based on Jasper
- ▶ Plugin and configuration compatibility with Nagios ®
- ▶ Public roadmap and active development

Architecture

- ▶ Built on Icinga Core
 - ▶ Classic UI uses native status.dat/objects.cache
- ▶ IDOUtils DB Backend
- ▶ Icinga Web
 - ▶ REST API
 - ▶ Used by Icinga Mobile
- ▶ Icinga Reporting
- ▶ Icinga QA, Testing & VM
- ▶ Changes in 2011
 - ▶ Icinga API as abstraction layer to IDOUtils DB deprecated
 - ▶ Icinga Web with Doctrine DB layer

Architecture



Icinga - Core

- ▶ Acknowledgement with expiry time
 - ▶ No fire & forget anymore
 - ▶ Does not affect reports
 - ▶ New commands
- ▶ Dualstacked IPv6 support
 - ▶ Host attribute address6
 - ▶ \$HOSTADDRESS6\$ macro
- ▶ Execute checks with execvp
 - ▶ Saves 1 fork
 - ▶ No shell meta characters -> fallback to popen

Icinga - Core

- ▶ Extended logging capabilities
 - ▶ Syslog facilities
 - ▶ Log long plugin output
 - ▶ Log errors on non existing objects in commands
- ▶ Module Object configuration
 - ▶ For event broker modules
 - ▶ No need to edit icinga.cfg – packager's wonderland
- ▶ Extended host & service stalking logic
 - ▶ Event handlers & notifications

Icinga - Core

- ▶ Handling of perfdata with empty results
 - ▶ Regression on addons & distributed setups
 - ▶ Fix denied in Nagios
- ▶ Enhanced performance
 - ▶ Check scheduling
 - ▶ Notifications
 - ▶ Macro processing
- ▶ Downtimes
 - ▶ Flexible downtimes now last trigger time+duration
 - ▶ Not resent on core restart anymore

Icinga - Classic UI

- ▶ Compound Commands
 - ▶ Acknowledge/Reschedule/etc multiple hosts or services at once
 - ▶ Delete multiple downtimes

Icinga 1.7.0 - Logged in as icingaadmin

▶ View **History** For All Hosts
 ▶ View **Notifications** For All Hosts
 ▶ View **Host AND Services** For All Hosts
 ▶ View **Host Status Detail** For All Hosts

Commands for checked services

Select command ▼

Submit

Display Filters: +

Service Status Details For All Hosts

Host ^v	Service ^v	Status ^v	Last Check ^v	Duration ^v	Attempt ^v	Status Information	
HOSTDNS	DNS Report	UNKNOWN	2012-05-31 19:19:21	0d 6h 7m 32s	1/1 (#0)	UNKNOWN: Report	☐
HOSTICINGA	All Disks	CRITICAL	2012-05-31 19:27:02	10d 23h 3m 23s	5/5 (#0)	DISK CRITICAL - free space: / 2335 MB (26% inode=62%): /lib/init/rw 505 MB (100% inode=99%): /dev 500 MB (99% inode=99%): /dev/shm 505 MB (100% inode=99%):	☐
	All Procs	CRITICAL	2012-05-31 19:25:37	10d 22h 21m 46s	5/5 (#0)	PROCS CRITICAL: 173 processes	☐
	PNP Stale RRDs	WARNING	2012-05-31 19:27:15	0d 6h 6m 31s	5/5 (#0)	WARNING: 649 XML Files checked. 0 RRD Errors found. 484 old XML Files found	☐

Icinga - Classic UI

- ▶ Revamped tactical overview
 - ▶ Don't show passive checks as disabled
 - ▶ use the signal-colour red more economically

Tactical Monitoring Overview

Last Updated: Thu May 31 19:50:55 CEST 2012 - Update in 65 seconds [pause] 
Icinga 1.7.0 - Logged in as icingaadmin

Network Outages

1 Outages

1 Blocking Outages

Hosts

2 Down

4 Unreachable

26 Up

1 Pending

Unhandled Problems
2 Active

Unhandled Problems
4 Active

Services

17 Critical

8 Warning

8 Unknown

122 Ok

5 Pending

Unhandled Problems
10 Active
1 Passive

Unhandled Problems
6 Active

Unhandled Problems
3 Active
1 Passive

Scheduled Downtime
4 Active

Scheduled Downtime
2 Active

Scheduled Downtime
1 Active

on Problem Hosts
Unacknowledged
1 Active
Scheduled Downtime
1 Active

on Problem Hosts
Unacknowledged
3 Active

Service Checks

Active

Passive

Enabled

146 Enabled

Enabled

14 Enabled

Host Checks

Active

Passive

Enabled

31 Enabled

Enabled

2 Enabled

Monitoring Features

Flap Detection

Notifications

Event Handlers

Enabled

2 Services Disabled
6 Services Flapping

Enabled

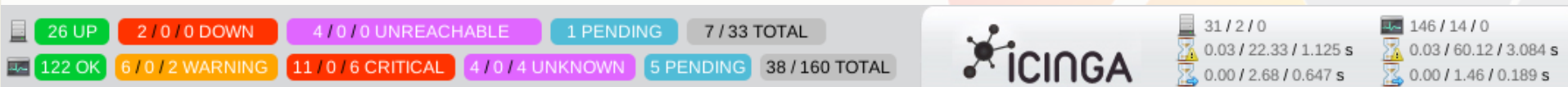
All Services Enabled
All Hosts Enabled

Enabled

11 Services Disabled
1 Host Disabled

Icinga - Classic UI

- ▶ New Status Header
 - ▶ Different states, ack/unack, pending, totals
 - ▶ URLs to extended filters
 - ▶ Autorefresh
 - ▶ Keep the problems in your views



Icinga - Classic UI

- ▶ Advanced RegEx search
 - ▶ Hosts AND services
- ▶ Advanced Filters
- ▶ Highlight table rows
- ▶ Page refresh with pause/continue
- ▶ Show comments as tooltip in status.cgi
- ▶ Advanced data export
 - ▶ JSON, CSV, XML
- ▶ Check for status data freshness
- ▶ Acknowledgements with opt-in expiry time

Icinga - Classic UI

27 UP

2 / 0 / 0 DOWN

4 / 0 / 0 UNREACHABLE

1 PENDING

7 / 34 TOTAL

127 OK

9 / 0 / 0 WARNING

12 / 0 / 2 CRITICAL

5 / 0 / 3 UNKNOWN

5 PENDING

36 / 163 TOTAL



32 / 2 / 0

0.04 / 10.49 / 0.534 s

0.00 / 19.16 / 0.772 s

149 / 14 / 0

0.03 / 60.23 / 3.656 s

0.00 / 37.23 / 1.112 s

General

Home

Documentation

Search: process*

Status

Tactical Overview

Host Detail

Service Detail

Hostgroup Overview

Hostgroup Summary

Servicegroup Overview

Servicegroup Summary

Status Map

Problems

Service Problems

Unhandled Services

Host Problems

Unhandled Hosts

All Unhandled Problems

Network Outages

System

Comments

Downtime

Process Info

Performance Info

Scheduling Overview

Current Network Status

Last Updated: Sun Jun 3 12:51:22 CEST 2012

Icinga 1.7.0 - Logged in as icingaadmin

Update in 76 seconds [pause]

View Host Status Detail For All Hosts

Commands for checked services

Select command

Submit

Service Status Details For Host/Services matching 'process*'

Host	Service	Status	Last Check	Duration	Attempt	Status Information
HOSTICINGA	icinga Process	OK	2012-06-03 12:50:59	13d 15h 24m 58s	1/5	PROCS OK: 7 processes with command name 'icinga'
	apache2 Process	OK	2012-06-03 12:50:57	13d 15h 24m 57s	1/5	PROCS OK: 11 processes with command name 'apache2'
	ido2db Process	OK	2012-06-03 12:50:58	1d 15h 29m 9s	1/5	PROCS OK: 2 processes with command name 'ido2db'
	mysql Process	OK				
	npd Process	OK				
localhost	Total Processes	OK				
	ido2db Process	OK	2012-06-01 21:29:14	2d 16h 52m 43s	1/4	PROCS OK: 2 processes with command name 'ido2db'

7 Matching Service Entries Displayed

Icinga - Classic UI

- ▶ Logfile search supported via webinterface
 - ▶ Extended filters
- ▶ Log commands (optional)
 - ▶ User, ip address, command
- ▶ Full command resolution in config.cgi
 - ▶ \$ARGn\$, \$USERn\$ + host/service macros
- ▶ Full display name support
 - ▶ Like host alias, set a better name for the GUI
- ▶ Multiple URLs for action | notes_url
- ▶ Acknowledgement of blocking outages

Icinga - Classic UI

Current Event Log

Last Updated: Sun Jun 3 12:48:37 CEST 2012
icinga 1.7.0 - Logged in as icingaadmin

Latest Archive
←

Log Navigation

Sun May 27 00:00:00 CEST 2012
to
Present..

Filters: 

Search:

Entries:

Order: ☒ Newer Entries First | ☐ Older Entries First

Timeperiod:

on off Type

☒ ☐ Notifications

☒ ☐ Host Status

☒ ☐ Service Status

☒ ☐ External Commands

☐ ☒ System Messages

☒ ☐ Event Handler

☒ ☐ Flapping

☒ ☐ Downtime

Entry Type:

June 03, 2012 12:00

 [2012-06-03 12:48:32] SERVICE ALERT: HOSTCHECKMK;NTP;OK;HARD;1;NTP OK: Offset -0.009846 secs
 [2012-06-03 12:48:12] SERVICE ALERT: HOSTICINGA;Syslog;CRITICAL;SOFT;3;CRITICAL - (1 errors in check_logfiles.protocol-2012-06-03-12-48-09) - Jun 3 12:47:12 tfnoc-icinga icinga: SERVICE ALERT: HOSTICINGA;Syslog;CRITICAL:SOFT;2;CRITICAL - (1 errors in check_logfiles.protocol-2012-06-03-12-47-09) - Jun 3 12:46:12 tfnoc-icinga icinga: SERVICE ALERT: HOSTICINGA;Syslog;CRITICAL:SOFT;1;CRITICAL - (1 errors in check_logfiles.protocol-2012-06-03-12-46-10) - Jun 3 12:46:02 tfnoc-icinga icinga: SERVICE ALERT: HOSTICINGA;Syslog;OK:HARD;5;OK - no errors or warnings
 [2012-06-03 12:47:47] HOST ALERT: host_11;DOWN;SOFT;1;host_11 (checked by tfnoc-icinga) FLAP: flap hostcheck down
 [2012-06-03 12:47:12] SERVICE ALERT: HOSTICINGA;Syslog;CRITICAL;SOFT;2;CRITICAL - (1 errors in check_logfiles.protocol-2012-06-03-12-47-09) - Jun 3 12:46:12 tfnoc-icinga icinga: SERVICE ALERT: HOSTICINGA;Syslog;CRITICAL:SOFT;1;CRITICAL - (1 errors in check_logfiles.protocol-2012-06-03-12-46-10) - Jun 3 12:46:02 tfnoc-icinga icinga: SERVICE ALERT: HOSTICINGA;Syslog;OK:HARD;5;OK - no errors or warnings
 [2012-06-03 12:47:02] SERVICE ALERT: host_13;random_0;WARNING;HARD;3;host_13 (checked by tfnoc-icinga) WARNING: random random_0 unchanged
 [2012-06-03 12:46:52] SERVICE ALERT: HOSTICINGA;APT;OK;SOFT;2;APT OK: 0 packages available for upgrade (0 critical updates).
 [2012-06-03 12:46:12] SERVICE ALERT: HOSTICINGA;Syslog;CRITICAL;SOFT;1;CRITICAL - (1 errors in check_logfiles.protocol-2012-06-03-12-46-10) - Jun 3 12:46:02 tfnoc-icinga icinga: SERVICE ALERT: HOSTICINGA;Syslog;OK:HARD;5;OK - no errors or warnings
 [2012-06-03 12:46:02] SERVICE ALERT: HOSTICINGA;APT;CRITICAL;SOFT;1;CRITICAL - Plugin timed out after 10 seconds
 [2012-06-03 12:46:02] SERVICE ALERT: HOSTICINGA;Syslog;OK:HARD;5;OK - no errors or warnings
 [2012-06-03 12:45:02] SERVICE ALERT: HOSTNSCA;5min-report;OK;HARD;1;cron ran ok, sending report ... 3 ... 2 ... 1 ... b00m!
 [2012-06-03 12:44:57] SERVICE ALERT: host_13;random_0;WARNING;SOFT;2;host_13 (checked by tfnoc-icinga) WARNING: random random_0 unchanged
 [2012-06-03 12:44:32] SERVICE ALERT: host_10;flap_0;OK;SOFT;2;host_10 (checked by tfnoc-icinga) FLAP: up flap_0 up
 [2012-06-03 12:44:22] SERVICE ALERT: host_08;flap_4;OK;SOFT;2;host_08 (checked by tfnoc-icinga) FLAP: up flap_4 up
 [2012-06-03 12:42:02] SERVICE ALERT: HOSTNRPE;NRPE Total Procs;WARNING;HARD;5;PROCS WARNING: 164 processes
 [2012-06-03 12:41:57] SERVICE ALERT: HOSTICINGA;HTTP Tomcat;OK;SOFT;2;HTTP OK: HTTP/1.1 200 OK - 8024 bytes in 0.010 second response time

Icinga - Classic UI

- ▶ Maintenance view for hosts & services in downtime
- ▶ Combined view of all unhandled hosts & services
- ▶ Show partial hostgroups opt-in
- ▶ Show notification number opt-in
- ▶ Show Scheduling Queue
- ▶ Show host/service dependencies, parent relations
- ▶ Various improvements
 - ▶ Navigation
 - ▶ Error messages
 - ▶ http charset option

Icinga - IDOUtils

- ▶ MySQL support
 - ▶ Rewritten to libdbi as db abstraction layer
 - ▶ Fixes for case insensitive columns
- ▶ Oracle support
 - ▶ Based on OCILIB
 - ▶ Prepared statements, bind params
 - ▶ MERGE statements emulating UPSERT
 - ▶ CLOBs for output, perfdata columns
 - ▶ Index tablespace
 - ▶ Advanced SQL setup script

Icinga - IDOUtils

- ▶ PostgreSQL support
 - ▶ Currently based on libdbi
 - ▶ Rewritten MySQL specific queries to UPDATE or INSERT
- ▶ Optimized data processing options
 - ▶ From users feedback, special defaults
- ▶ Optimized performance
 - ▶ Only insert finished checks
- ▶ Timezone awareness using UTC
 - ▶ Distributed nodes all over the world
 - ▶ Application shows correct timestamps

Icinga - IDOUtils

- ▶ Extended Logging
 - ▶ Syslog
 - ▶ Debug log for ido2db and idomod
- ▶ Add db socket as config option
- ▶ Automatic db version check
 - ▶ For upgrade safety
- ▶ Various improvements in housekeeping
 - ▶ Trim logentries, acks, notifications

Icinga - Documentation

- ▶ Documentation
 - ▶ Docbook in English & German
 - ▶ Docs for new features
 - ▶ Docs from feedback of testing & installing
 - ▶▶ Command list, extended CGI params, etc
 - ▶ PNP as default graphing solution
- ▶ Wiki
 - ▶ Installation guides for various platforms
 - ▶ Addon documentation- and installations-guides
 - ▶ Developer's docs space

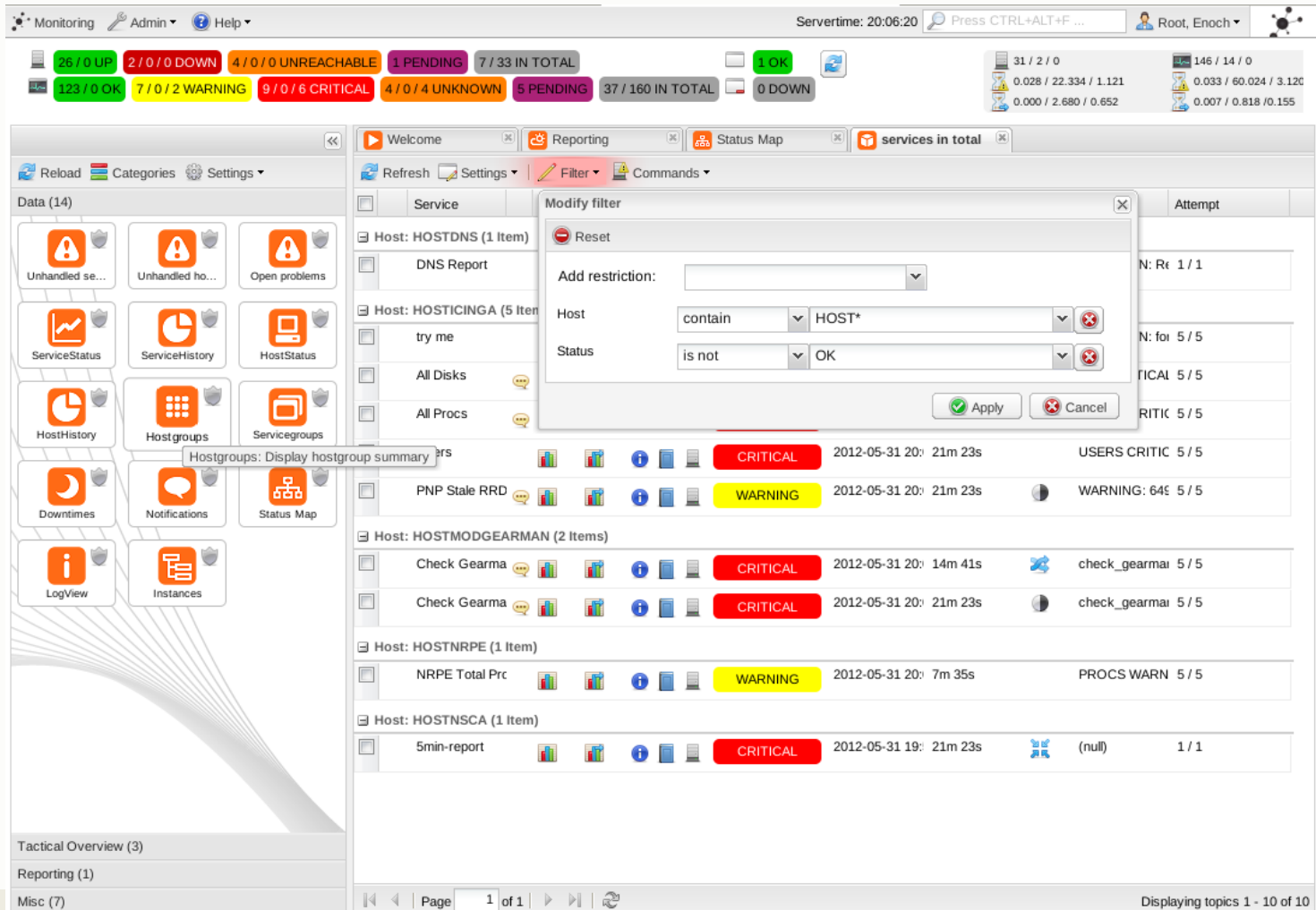
Icinga - Web

- ▶ Developed from scratch
- ▶ Based on Agavi MVC & Sencha (former EXTJS)
- ▶ IDOUtils RDBMS as datasource
 - ▶ Therefore support for MySQL, Postgresql, Oracle
 - ▶ Doctrine based abstraction layer supports various databases with easier configuration
- ▶ Cronks == widgets
- ▶ Dynamic dashboards
- ▶ Live search

Icinga - Web

- ▶ Sort and group data by columns
- ▶ Filter data in detail
 - ▶ E.g. all services critical by instance
 - ▶ Type and get live proposed results
- ▶ Send commands simultaneously to multiple hosts/services by checking boxes
- ▶ View host or service details and history in the same window
- ▶ Extended object information box
 - ▶ Status, relations, references

Icinga - Web



The screenshot displays the Icinga Web interface. At the top, a status bar shows server time (20:06:20) and user (Root, Enoch). Below this, a summary bar indicates overall system health: 26/0 UP, 2/0/0 DOWN, 4/0/0 UNREACHABLE, 1 PENDING, 7/33 IN TOTAL, 1 OK, 146/14/0, 123/0 OK, 7/0/2 WARNING, 9/0/6 CRITICAL, 4/0/4 UNKNOWN, 5 PENDING, 37/160 IN TOTAL, 0 DOWN.

The left sidebar contains navigation icons for various monitoring components: Unhandled services, Unhandled hosts, Open problems, ServiceStatus, ServiceHistory, HostStatus, HostHistory, Hostgroups, Servicegroups, Downtimes, Notifications, Status Map, LogView, and Instances.

The main content area shows a list of hosts and their services. A 'Modify filter' dialog box is open, allowing users to refine the view. The dialog includes a 'Reset' button, an 'Add restriction' field, and a table with columns for 'Host' and 'Status'. The table shows the following restrictions:

Host	Status
contain	HOST*
is not	OK

The main content area also displays a list of hosts and their services, including 'HOSTDNS (1 Item)', 'HOSTICINGA (5 Items)', 'HOSTMODGEARMAN (2 Items)', 'HOSTNRPE (1 Item)', and 'HOSTNSCA (1 Item)'. Each host entry shows its status (e.g., CRITICAL, WARNING) and a list of services with their respective statuses.

Icinga - Web

- ▶ Your own views – Cronks can be
 - ▶ Drag & dropped
 - ▶ Resized & minimized
 - ▶ Customized & created
 - ▶ Saved & exported & shared
- ▶ Multi column layout
 - ▶ Combine multiple cronks into one
- ▶ Fullscreen support
- ▶ Refresh for Cronks
- ▶ Tab slider like Cacti

Icinga - Web

Monitoring Admin Help Servertime: 20:09:20 Press CTRL+ALT+F ... Root, Enoch 1

26 / 0 UP 2 / 0 / 0 DOWN 4 / 0 / 0 UNREACHABLE 1 PENDING 7 / 33 IN TOTAL 1 OK 31 / 2 / 0 146 / 14 / 0
 121 / 0 OK 7 / 0 / 2 WARNING 12 / 0 / 5 CRITICAL 4 / 0 / 4 UNKNOWN 5 PENDING 39 / 160 IN TOTAL 0 DOWN 0.028 / 22.334 / 1.137 0.033 / 60.059 / 3.046
 0.000 / 2.680 / 0.651 0.002 / 0.777 / 0.143

Welcome Reporting services in total 3 Portal 3columns

Open problems

Refresh Settings Filter

Host	Service	Host Status	Service state
hos		UNREACHABLE	
hos		DOWN	
hos		UNREACHABLE	
hos		UNREACHABLE	
hos		UNREACHABLE	
hos		DOWN	
hos	bloc	UNREACHABLE	UNKNOWN
hos	criti	DOWN	CRITICAL
hos	bloc	UP	UNKNOWN
hos	criti	UP	CRITICAL
hos	rami	UP	WARNING
hos	bloc	UNREACHABLE	UNKNOWN

Page 1 of 2 Displaying topics 1 - 2

Downtimes

Refresh Settings Filter Commands

Obj	Host	Service status	Dow	Scheduled start	Sch
loc		OK		2012-05-18 13:51:35	20
HO		OK		2012-05-31 19:49:41	20
hos		WARNING		2012-05-31 19:49:41	20
HO		WARNING		2012-05-31 19:49:41	20
hos		CRITICAL		2012-05-31 19:49:41	20
HO		CRITICAL		2012-05-31 19:49:41	20
HO		CRITICAL		2012-05-31 19:49:41	20
HO		CRITICAL		2012-05-31 19:49:41	20
hos		UNKNOWN		2012-05-31 19:49:41	20

Page 1 of 1 Displaying topics 1 - 9

LogView

Refresh Settings Filter

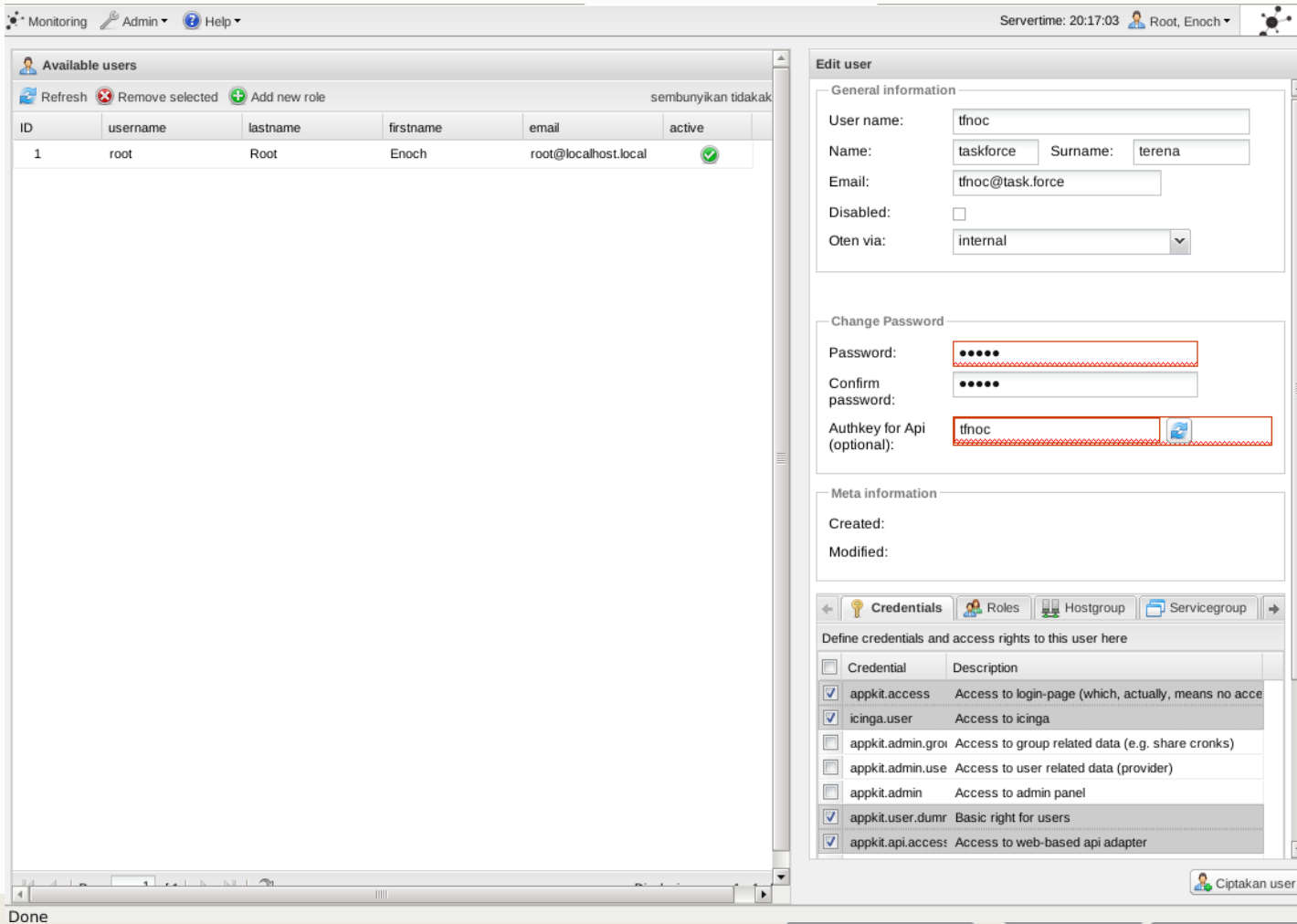
Timestamp	Type	Message
Hour: 2012-05-31 20:00:00 (25 Items)		
2012-05-31 20:09:13	service OK	SERVICE ALERT: host_16:fl (checked by icinga-demo-deb FLAP: up flap_2 up
2012-05-31 20:08:53	passive check	PASSIVE SERVICE CHECK: HOSTCHI - 69.2% used (6.17 of 8.9 GB), (levels at 80.0/90.0) trend: +38.43MB / 24 hours
2012-05-31 20:08:53	passive check	PASSIVE SERVICE CHECK:

Page 1

Icinga - Web

- ▶ User management
 - ▶ Default contacts & contactgroups layout
 - ▶ Extended permissions based on host/servicegroups
 - ▶ Limit to custom variables set
 - ▶ Easy administration of complex access rules
- ▶ Auth provider
 - ▶ Internal
 - ▶ HTTP auth
 - ▶ LDAP
 - ▶ AD

Icinga Web



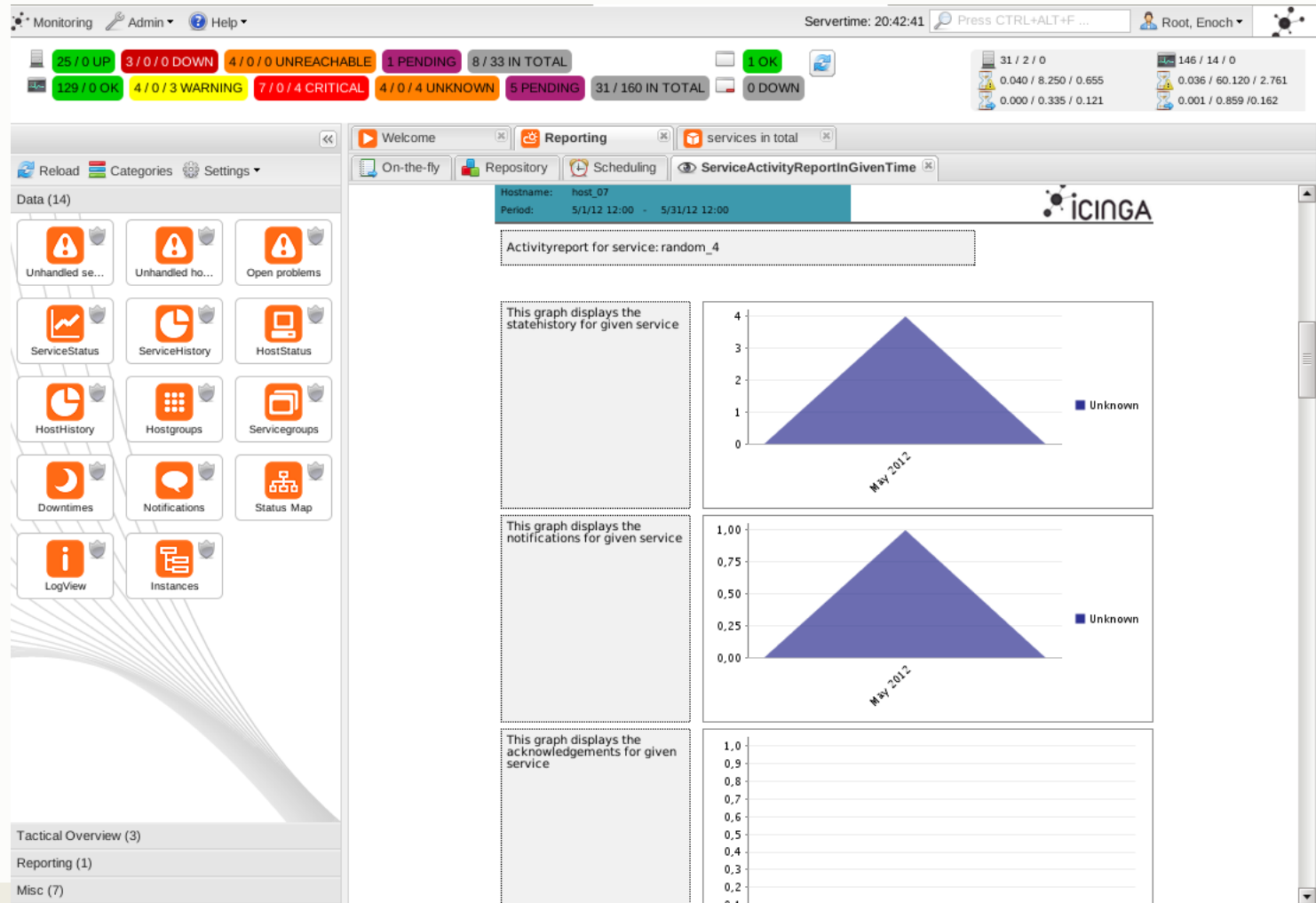
The screenshot shows the Icinga Web Admin interface. The top navigation bar includes 'Monitoring', 'Admin', and 'Help'. The right side shows the server time '20:17:03' and the user 'Root, Enoch'. The main content area is divided into two panels. The left panel, titled 'Available users', shows a table with one user: 'root' (ID 1, username 'root', lastname 'Root', firstname 'Enoch', email 'root@localhost.local', active). The right panel, titled 'Edit user', shows the 'General information' section with fields for 'User name' (tfnoc), 'Name' (taskforce), 'Surname' (terena), 'Email' (tfnoc@task.force), 'Disabled' (checkbox), and 'Oten via' (internal). Below this is the 'Change Password' section with fields for 'Password', 'Confirm password', and 'Authkey for Api (optional)'. The 'Meta information' section shows 'Created' and 'Modified' fields. At the bottom, there is a 'Credentials' tab with a table of access rights for the user 'tfnoc'.

Credential	Description
☒ appkit.access	Access to login-page (which, actually, means no acce
☒ icinga.user	Access to icinga
☐ appkit.admin.groi	Access to group related data (e.g. share crons)
☐ appkit.admin.use	Access to user related data (provider)
☐ appkit.admin	Access to admin panel
☒ appkit.user.dumr	Basic right for users
☒ appkit.api.access	Access to web-based api adapter

Icinga - Web

- ▶ HTTP / REST API
 - ▶ Authkey protected
 - ▶ Easing addon development
 - ▶ Used by Icinga Mobile
- ▶ Reporting integration
 - ▶ Execute reports in Icinga Web
 - ▶ Download supported formats directly without access to Jasper
 - ▶ Job scheduling

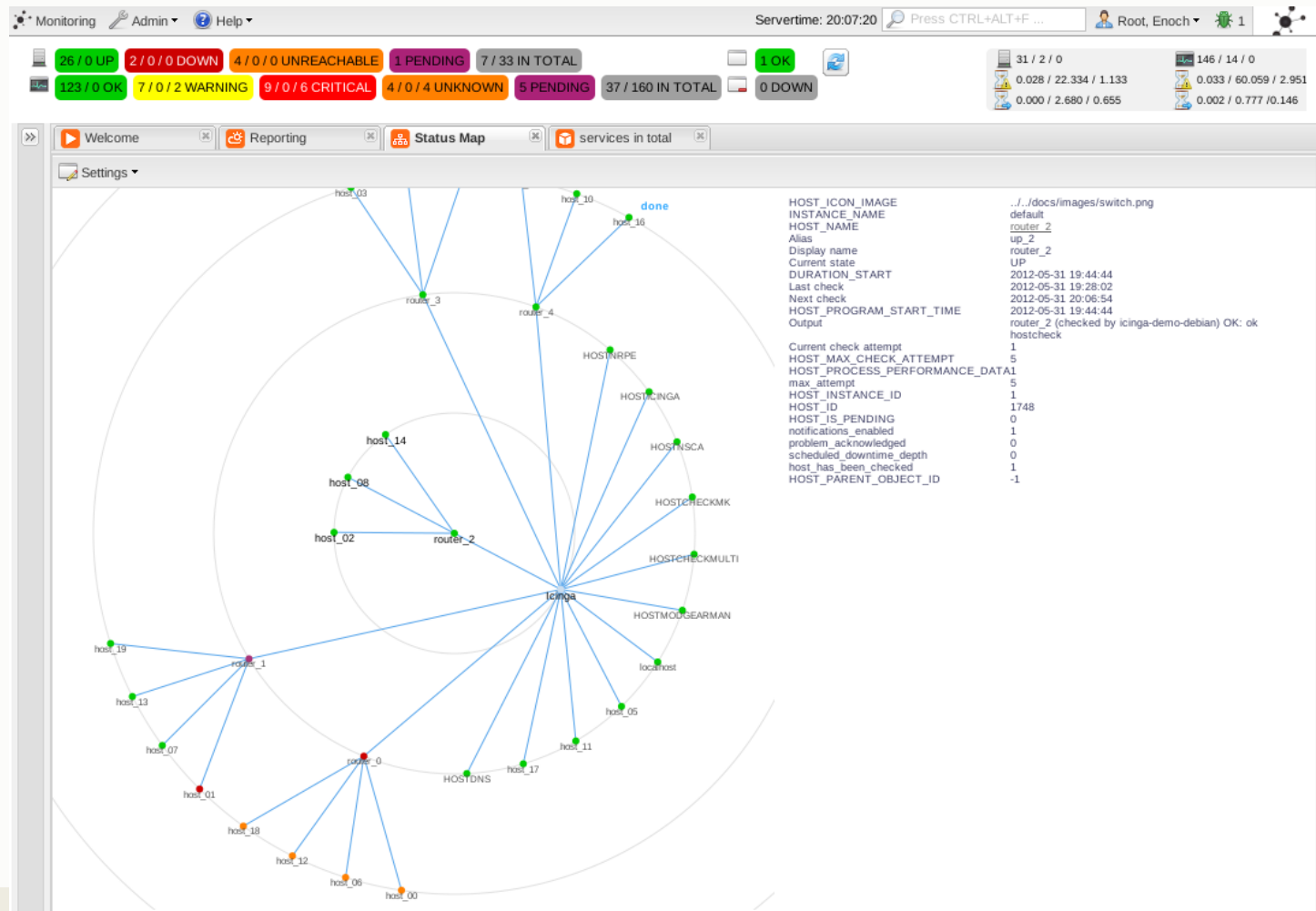
Icinga - Web



Icinga - Web

- ▶ New module interface
 - ▶ Update safe modules
 - ▶ Independent module configuration and installation
- ▶ Statusmap
 - ▶ Scalable & Zoomable
 - ▶ Animated Centering
- ▶ Integration of Addons
 - ▶ PNP (check contrib)
 - ▶ Business Process Cronk
 - ▶ Lconf Config Editor Cronk
 - ▶ inGraph

Icinga - Web



Icinga - Mobile

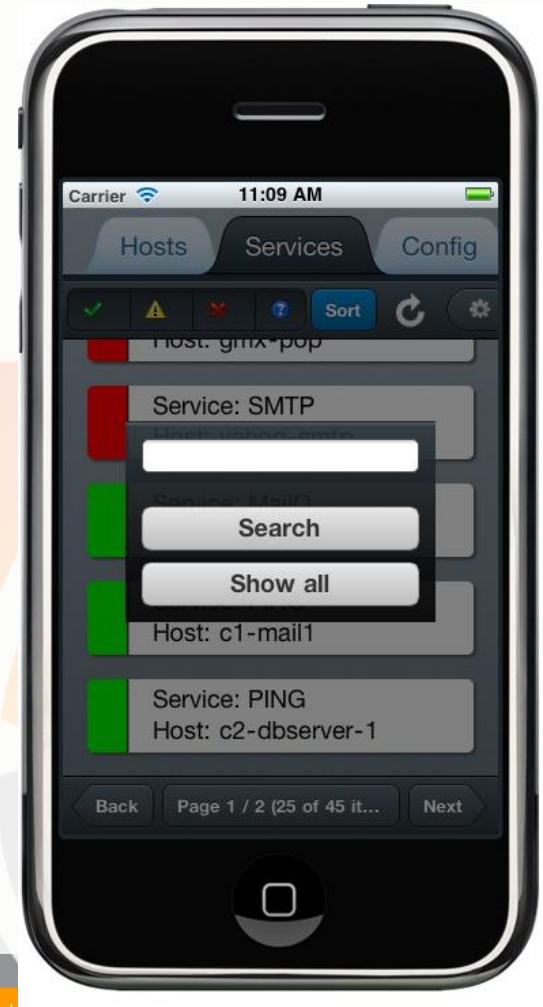
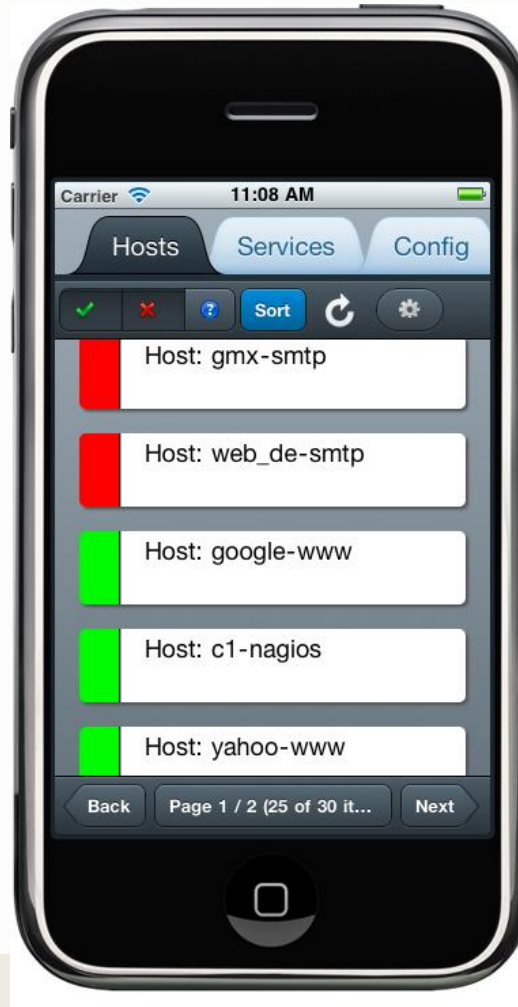
► Facts

- Written in JavaScript
- Enriched with Sencha Touch
- Installation free – just add to phone bookmarks
- Uses Icinga Web REST API

► Features

- Authentication via API key applies all Icinga Web user rights and preferences
- Local storage of settings on phone
- Central updates apply automatically to all Icinga Mobile users

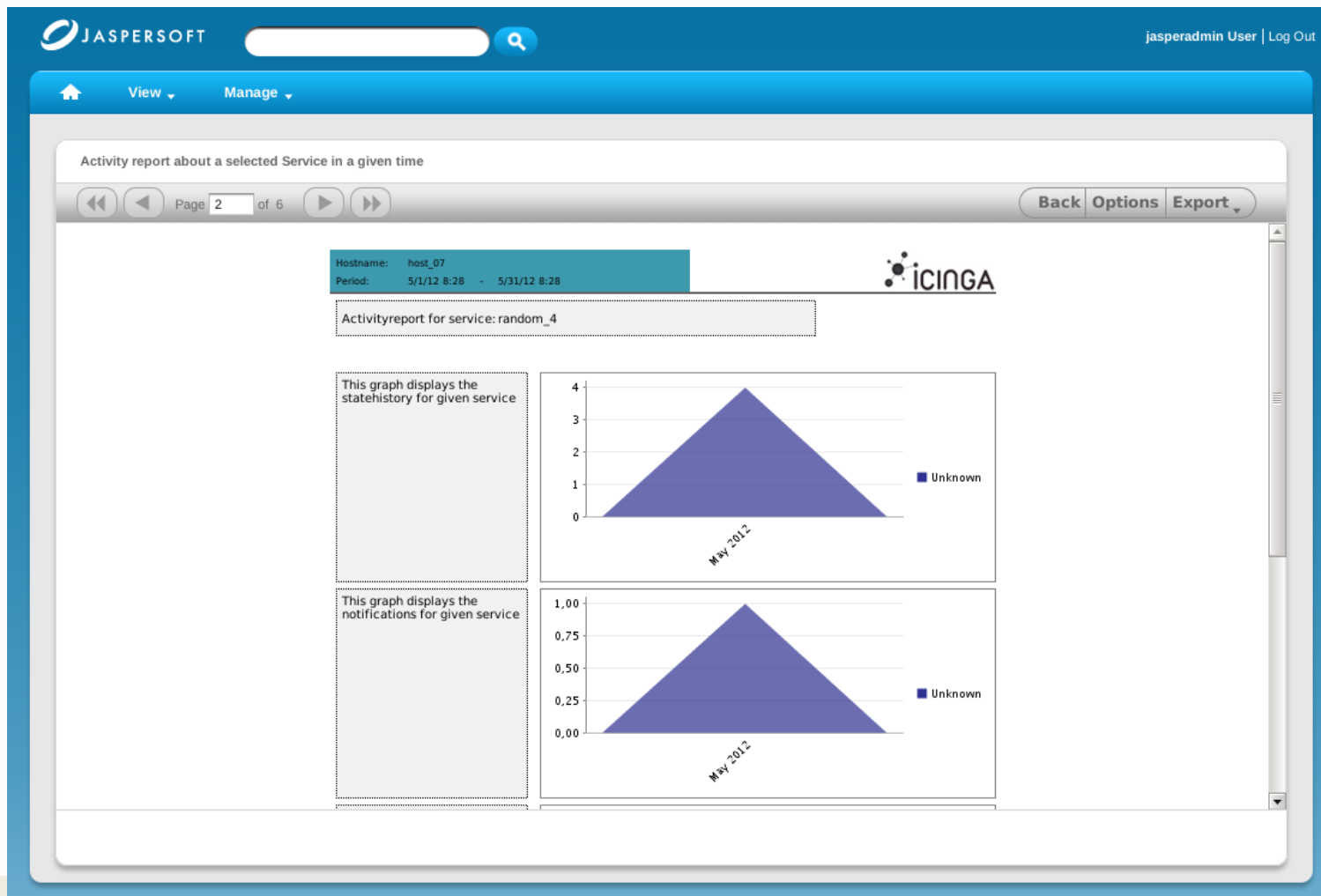
Icinga - Mobile



Icinga - Reporting

- ▶ Based on Jasper Reports
- ▶ Various Report Templates included
 - ▶ Synopsis report of the complete monitoring environment
 - ▶ Top 10 problematic hosts or services report
 - ▶ Groups, hosts or services availability reports
 - ▶ Acknowledgement records
- ▶ SOAP interface used in Icinga Web
 - ▶ Add your own app!
- ▶ Create your own reports with iReport

Icinga - Reporting



Icinga Addons & Plugins

- ▶ Graphing
 - ▶ PNP4Nagios, inGraph
- ▶ Transport & Interfaces
 - ▶ check_multi, mod_gearman
 - ▶ NRPE, NSCA, NSClient++, check_mk
- ▶ Configuration & Management
 - ▶ Nagiosql, NConf, Lconf
 - ▶ NoMa
 - ▶ Puppet, Chef
- ▶ GUI & Visualization
 - ▶ NagVis, NagStamon, aNag

Icinga Addons & Plugins

▶ Plugins

- ▶ Nagios Plugins still work ☺
- ▶ <https://wiki.icinga.org/display/howtos/Plugins>
- ▶ Your own ...

▶ Plugin Collections

- ▶ <https://www.monitoringexchange.org>
- ▶ <https://wiki.icinga.org/display/howtos/Plugin+Collections>

Icinga Packages

- ▶ Enhancements
 - ▶ Many flags and changes to make packaging easy
 - ▶ Spec file provided
- ▶ Packagers join Team Icinga
 - ▶ Pre-package builds on release
 - ▶ Easier communication on changes
 - ▶ <https://www.icinga.org/download/packages/>
- ▶ Icinga integrated into OMD
 - ▶ Core & Classic UI
 - ▶ No Database support in OMD -> no Icinga Web

Icinga Packages

› Debian & Ubuntu

- › Alexander Wirt
- › Stable & Debian Backports
- › Ubuntu merges even versions from testing
- › Package Build Server
 - › <http://icingabuild.dus.dg-i.net/>
 - › Icinga Web packages in progress

› RHEL & CentOS

- › Repoforge / Michael Friedrich
 - › Icinga, Icinga Web (soon)
- › EPEL – working on it
 - › https://bugzilla.redhat.com/show_bug.cgi?id=693608

Icinga Packages

- ▶ SuSe
 - ▶ Tim Hardeck
 - ▶ Icinga and Icinga Web
- ▶ FreeBSD
 - ▶ Lars Engels
 - ▶ Icinga, Web in progress
- ▶ Gentoo
 - ▶ Matthew Thode
 - ▶ Icinga and Icinga Web
- ▶ Solaris, OpenBSD, Archlinux
 - ▶ Check icinga.org

Roadmap

- ▶ Planned public events
 - ▶ Open Source Monitoring Conference 2012
- ▶ Versions planned for 2012
 - ▶ 05/15/2012 – 1.7
 - ▶ 10/17/2012 – 1.8

What's next ?

- ▶ Maintenance Release 1.7
- ▶ Feature Release 1.8
 - ▶ Extend Reporting to Postgresql/Oracle
- ▶ Performance improvements
 - ▶ Optimized check result handling
- ▶ Optimized installation process
 - ▶ Upstream Packages
- ▶ Prototype Message Queuing Systems
 - ▶ APIs, Message Bus Systems

Questions & Answers



Questions & Answers

Web www.icinga.org

Development dev.icinga.org

Feedback feedback.icinga.org

Wiki wiki.icinga.org

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Everywhere!